

MH1812 Discrete Mathematics

– Continuous Assessment 2

NTU, AY18/19 S1

1st Nov 2018, 10:30AM - 11:30AM @ LKC-LT(MLT)

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Tutorial Group:

Note: first 5 questions are compulsory. Question 6 is bonus, you get the points if you answer it correctly, you don't lose any points if you leave it or answer wrongly. Write on the back of paper if there is not enough space. Leave the tutorial group if you can't remember.

Question 1. (20 points)

Given the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2}$, with $a_1 = 6$ and $a_0 = 2$, find the explicit formula of a_n .

$$5'' \quad x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$3'' \quad x = 2$$

$$2'' \quad a_n = u \cdot 2^n + v \cdot n \cdot 2^n$$

$$3'' \quad a_n = 2^{n+1} + n \cdot 2^n$$

$$2'' \quad 6 = a_1 = u \cdot 2 + v \cdot 2$$

$$2'' \quad 2 = a_0 = u$$

$$\left. \begin{matrix} 6 = 2u + 2v \\ 2 = u \end{matrix} \right\} \Rightarrow u = 2, v = 1 \quad 3''$$

Question 2. (10 points)

Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, at least how many numbers does one need to select from A to be sure there exist two selected numbers whose sum is even.

3

10''

Question 3. (30 points)

Define on set $A = \{a, b, c, d\}$ the binary relation $R = \{(a, b), (b, a), (c, d), (d, c)\}$.

1. Is R reflexive, symmetric, transitive, anti-symmetric?
2. find the transitive closure R^t .

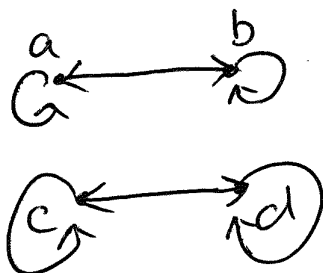
1.4" a. reflexive $\times$ $(a, a) \notin R$

4" b. symmetric $\checkmark$

4" c. transitive $\times$ $(a, b), (b, a) \in R, (a, a) \notin R$

4" d. Anti-symmetric $\times$ $(a, b), (b, a) \in R$

2.



$$R^t = \left\{ \begin{array}{l} (a, a), (a, b), (b, a), (b, b) \\ (c, c), (c, d), (d, c), (d, d) \end{array} \right\}$$

total 14"

miss^o the original terms in R (-7)

others (-3) for each wrongly added / missing term.

Question 4. (20 points)

1. Let set $A = \{a, b\}$, what is the power set $P(A)$.
2. Let set $B = \{b, c\}$, what are $A \cup B, A \cap B, A - B$?

1. $P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

8" (-3) for each missing term

(-3) for wrong notation (once only)

2. 4" $A \cup B = \{a, b, c\}$

4" $A \cap B = \{b\}$

4" $A - B = \{a\}$

NO working mark.

only "0" mark or "4" for each.

Question 5. (20 points)

Let $f, g: \mathbb{Z} \rightarrow \mathbb{Z}$, with $f(x) = x^2 - 2$ and $g(x) = x + 1$,

1. find g^{-1} .

2. find $f \circ g$.

1. 8" $y = x + 1$

$x = y - 1 \leftarrow 4"$

$g^{-1}(x) = x - 1 \leftarrow 4"$

2. 12" $f \circ g(x) = f(x + 1) \leftarrow 6"$

$= (x + 1)^2 - 2 \leftarrow 6"$

3

if calculated $g \circ f(x) = x^2 - 2 + 1 = x^2 - 1$

6" (Any mistake 0")

Bonus Question 6 (10 points)

Defined on a set of size n , how many anti-symmetric relations are there?

$$A = \{a_1, a_2, a_3 \dots a_n\}$$

$$S = \{ \underline{(a_1, a_1), (a_2, a_2) \dots (a_n, a_n)} \}$$

4"

2^n choices here.

For each (a_i, a_j) with $a_i \neq a_j$

Anti-Symmetric accepts three cases:

$$3'' \left\{ \begin{array}{l} \emptyset, \\ (a_i, a_j) \in R \wedge (a_j, a_i) \notin R \\ (a_j, a_i) \in R \wedge (a_i, a_j) \notin R \end{array} \right.$$

2'' There are $\binom{n}{2}$ such pairs

~~2~~ $3^{\binom{n}{2}}$ choices

$$1'' \quad 2^n \cdot 3^{\binom{n}{2}} \#$$